

Tables of Integral Transforms

In this chapter we provide a set of *short* tables of integral transforms of the functions that are either cited in the text or in most common use in mathematical, physical, and engineering applications. For exhaustive lists of integral transforms, the reader is referred to Erdélyi et al. (1954), Campbell and Foster (1948), Ditkin and Prudnikov (1965), Doetsch (1970), Marichev (1983), Debnath (1995), and Oberhettinger (1972).

13.1 Fourier Transforms

	$f(x)$	$F(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp(-ikx) f(x) dx$
1	$\exp(-a x), \quad a > 0$	$\left(\sqrt{\frac{2}{\pi}}\right) a (a^2 + k^2)^{-1}$
2	$x \exp(-a x), \quad a > 0$	$\left(\sqrt{\frac{2}{\pi}}\right) (-2aik) (a^2 + k^2)^{-2}$
3	$\exp(-ax^2), \quad a > 0$	$\frac{1}{\sqrt{2a}} \exp\left(-\frac{k^2}{4a}\right)$
4	$(x^2 + a^2)^{-1}, \quad a > 0$	$\sqrt{\frac{\pi}{2}} \frac{\exp(-a k)}{a}$
5	$x(x^2 + a^2)^{-1}$	$\sqrt{\frac{\pi}{2}} \left(\frac{ik}{2a}\right) \exp(-a k)$

	$f(x)$	$F(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp(-ikx) f(x) dx$
6	$\begin{cases} c, & a \leq x \leq b \\ 0, & \text{outside} \end{cases}$	$\frac{ic}{\sqrt{2\pi}} \frac{1}{k} (e^{-ibk} - e^{-iak})$
7	$ x \exp(-a x), \quad a > 0$	$\sqrt{\frac{2}{\pi}} (a^2 - k^2) (a^2 + k^2)^{-2}$
8	$\frac{\sin ax}{x}$	$\sqrt{\frac{\pi}{2}} H(a - k)$
9	$\exp\{-x(a - i\omega)\} H(x)$	$\frac{1}{\sqrt{2\pi}} \frac{i}{(\omega - k + ia)}$
10	$(a^2 - x^2)^{-\frac{1}{2}} H(a - x)$	$\sqrt{\frac{\pi}{2}} J_0(ak)$
11	$\frac{\sin b(x^2 + a^2)^{\frac{1}{2}}}{(x^2 + a^2)^{\frac{1}{2}}}$	$\sqrt{\frac{\pi}{2}} J_0(a\sqrt{b^2 - k^2}) H(b - k)$
12	$\frac{\cos(b\sqrt{a^2 - x^2})}{(a^2 - x^2)^{\frac{1}{2}}} H(a - x)$	$\sqrt{\frac{\pi}{2}} J_0(a\sqrt{b^2 + k^2})$
13	$e^{-ax} H(x), \quad a > 0$	$\frac{1}{\sqrt{2\pi}} (a - ik) (a^2 + k^2)^{-1}$
14	$\frac{1}{\sqrt{ x }} \exp(-a x), \quad a > 0$	$(a^2 + k^2)^{-\frac{1}{2}} \left[a + (a^2 + k^2)^{\frac{1}{2}} \right]^{\frac{1}{2}}$
15	$\delta^{(n)}(x - a), \quad n = 0, 1, 2, \dots$	$\frac{1}{\sqrt{2\pi}} (ik)^n \exp(-iak)$
16	$\exp(iax)$	$\sqrt{2\pi} \delta(k - a)$

13.2 Fourier Sine Transforms

	$f(x)$	$F_s(k) = \sqrt{\frac{2}{\pi}} \int_0^\infty \sin(kx) f(x) dx$
1	$\exp(-ax), \quad a > 0$	$\sqrt{\frac{2}{\pi}} k (a^2 + k^2)^{-1}$
2	$x \exp(-ax), \quad a > 0$	$\sqrt{\frac{2}{\pi}} (2ak) (a^2 + k^2)^{-2}$
3	$x^{\alpha-1}, \quad 0 < \alpha < 1$	$\sqrt{\frac{2}{\pi}} k^{-\alpha} \Gamma(\alpha) \sin\left(\frac{\pi\alpha}{2}\right)$
4	$\frac{1}{\sqrt{x}}$	$\frac{1}{\sqrt{k}}, \quad k > 0$
5	$x^{\alpha-1} e^{-ax}, \quad \alpha > -1, \quad a > 0$	$\sqrt{\frac{2}{\pi}} \Gamma(\alpha) r^{-\alpha} \sin(\alpha\theta), \text{ where}$ $r = (a^2 + k^2)^{\frac{1}{2}}, \quad \theta = \tan^{-1}\left(\frac{k}{a}\right)$
6	$x^{-1} e^{-ax}, \quad a > 0$	$\sqrt{\frac{2}{\pi}} \tan^{-1}\left(\frac{k}{a}\right), \quad k > 0$
7	$x \exp(-a^2 x^2)$	$2^{-3/2} \left(\frac{k}{a^3}\right) \exp\left(-\frac{k^2}{4a^2}\right)$
8	$\operatorname{erfc}(ax)$	$\sqrt{\frac{2}{\pi}} \frac{1}{k} \left[1 - \exp\left(-\frac{k^2}{4a^2}\right)\right]$
9	$x (a^2 + x^2)^{-1}$	$\sqrt{\frac{\pi}{2}} \exp(-ak), \quad a > 0$
10	$x (a^2 + x^2)^{-2}$	$\frac{1}{\sqrt{2\pi}} \left(\frac{k}{a}\right) \exp(-ak), \quad (a > 0)$

	$f(x)$	$F_s(k) = \sqrt{\frac{2}{\pi}} \int_0^\infty \sin(kx) f(x) dx$
11	$H(a-x), \quad a > 0$	$\sqrt{\frac{2}{\pi}} \frac{1}{k} (1 - \cos ak)$
12	$x^{-1} J_0(ax)$	$\left\{ \begin{array}{l} \sqrt{\frac{2}{\pi}} \sin^{-1}\left(\frac{k}{a}\right), \quad 0 < k < a \\ \sqrt{\frac{\pi}{2}}, \quad a < k < \infty \end{array} \right\}$
13	$x(a^2 + x^2)^{-1} J_0(bx),$ $a > 0, b > 0$	$\sqrt{\frac{\pi}{2}} e^{-ak} I_0(ab), \quad a < k < \infty$
14	$J_0(a\sqrt{x}), \quad a > 0$	$\sqrt{\frac{2}{\pi}} \frac{1}{k} \cos\left(\frac{a^2}{4k}\right)$
15	$(x^2 - a^2)^{\nu - \frac{1}{2}} H(x - a),$ $ \nu < \frac{1}{2}$	$2^{\nu - \frac{1}{2}} \left(\frac{a}{k}\right)^\nu \Gamma\left(\nu + \frac{1}{2}\right) J_{-\nu}(ak)$
16	$x^{1-\nu} (x^2 + a^2)^{-1} J_\nu(ax),$ $\nu - \frac{3}{2}, \quad a, b > 0$	$\sqrt{\frac{\pi}{2}} a^{-\nu} \exp(-ak) I_\nu(ab),$ $a < k < \infty$
17	$x^{-\nu} J_{\nu+1}(ax), \quad \nu > -\frac{1}{2}$	$\frac{k(a^2 - k^2)^{\nu - \frac{1}{2}}}{2^{\nu - \frac{1}{2}} a^{\nu+1} \Gamma(\nu + \frac{1}{2})} H(a - k)$
18	$\operatorname{erfc}(ax)$	$\sqrt{\frac{2}{\pi}} \frac{1}{k} \left[1 - \exp\left(-\frac{k^2}{4a^2}\right)\right]$
19	$x^{-\alpha}, \quad 0 < \alpha < 2$	$\Gamma(1 - \alpha) k^{\alpha-1} \cos\left(\frac{\alpha\pi}{2}\right)$
20	$(ax - x^2)^{\alpha - \frac{1}{2}} H(a - x),$ $\alpha > -\frac{1}{2}$	$\sqrt{2} \Gamma\left(\alpha + \frac{1}{2}\right) \left(\frac{a}{k}\right)^\alpha \sin\left(\frac{ak}{2}\right) J_\alpha\left(\frac{ak}{2}\right)$

13.3 Fourier Cosine Transforms

	$f(x)$	$F_c(k) = \sqrt{\frac{2}{\pi}} \int_0^\infty \cos(kx) f(x) dx$
1	$\exp(-ax), \quad a > 0$	$\left(\sqrt{\frac{2}{\pi}}\right) a (a^2 + k^2)^{-1}$
2	$x \exp(-ax), \quad a > 0$	$\left(\sqrt{\frac{2}{\pi}}\right) (a^2 - k^2) (a^2 + k^2)^{-2}$
3	$\exp(-a^2 x^2)$	$\frac{1}{a\sqrt{2}} \exp\left(-\frac{k^2}{4a^2}\right)$
4	$H(a - x)$	$\sqrt{\frac{2}{\pi}} \left(\frac{\sin ak}{k}\right)$
5	$x^{a-1}, \quad 0 < a < 1$	$\sqrt{\frac{2}{\pi}} \Gamma(a) k^{-a} \cos\left(\frac{a\pi}{2}\right)$
6	$\cos(ax^2)$	$\frac{1}{2\sqrt{a}} \left[\cos\left(\frac{k^2}{4a}\right) + \sin\left(\frac{k^2}{4a}\right)\right]$
7	$\sin(ax^2), \quad a > 0$	$\frac{1}{2\sqrt{a}} \left[\cos\left(\frac{k^2}{4a}\right) - \sin\left(\frac{k^2}{4a}\right)\right]$
8	$(a^2 - x^2)^{\nu-\frac{1}{2}} H(a - x), \quad \nu > -\frac{1}{2}$	$2^{\nu-\frac{1}{2}} \Gamma\left(\nu + \frac{1}{2}\right) \left(\frac{a}{k}\right)^\nu J_\nu(ak)$
9	$(a^2 + x^2)^{-1} J_0(bx), \quad a, b > 0$	$\sqrt{\frac{\pi}{2}} a^{-1} \exp(-ak) I_0(ab),$ $b < k < \infty$
10	$x^{-\nu} J_\nu(ax), \quad \nu > -\frac{1}{2}$	$\frac{(a^2 - k^2)^{\nu-\frac{1}{2}} H(a-k)}{2^{\nu-\frac{1}{2}} a^\nu \Gamma\left(\nu + \frac{1}{2}\right)}$

	$f(x)$	$F_c(k) = \sqrt{\frac{2}{\pi}} \int_0^\infty \cos(kx) f(x) dx$
11	$(x^2 + a^2)^{-\frac{1}{2}} \exp \left[-b(x^2 + a^2)^{\frac{1}{2}} \right]$	$K_0 \left[a(k^2 + b^2)^{\frac{1}{2}} \right], a > 0, b > 0$
12	$x^{\nu-1} e^{-ax}, \quad \nu > 0, a > 0$	$\sqrt{\frac{2}{\pi}} \Gamma(\nu) r^{-\nu} \cos n\theta$, where $r = (a^2 + k^2)^{\frac{1}{2}}, \theta = \tan^{-1} \left(\frac{k}{a} \right)$
13	$\frac{2}{x} e^{-x} \sin x$	$\sqrt{\frac{2}{\pi}} \tan^{-1} \left(\frac{2}{k^2} \right)$
14	$\sin \left[a(b^2 - x^2)^{\frac{1}{2}} H(b - x) \right]$	$\sqrt{\frac{\pi}{2}} (ab) (a^2 + k^2)^{-\frac{1}{2}} \times J_1 \left[b(a^2 + k^2)^{\frac{1}{2}} \right]$
15	$\frac{(1-x^2)}{(1+x^2)^2}$	$\sqrt{\frac{\pi}{2}} k \exp(-k)$
16	$x^{-\alpha}, \quad 0 < \alpha < 1$	$\sqrt{\frac{\pi}{2}} \frac{k^{\alpha-1}}{\Gamma(\alpha)} \sec \left(\frac{\pi\alpha}{2} \right)$
17	$\left(\frac{1}{a} + x \right) e^{-ax}$	$\sqrt{\frac{\pi}{2}} \frac{2a^2}{(a^2 + k^2)^2}$
18	$\log \left(1 + \frac{a^2}{k^2} \right), \quad a > 0$	$\sqrt{2\pi} \frac{(1 - e^{-ak})}{k}$
19	$\log \left(\frac{a^2 + x^2}{b^2 + x^2} \right), \quad a, b > 0$	$\sqrt{2\pi} \frac{(e^{-bk} - e^{-ak})}{k}$
20	$a(x^2 + a^2)^{-1}, \quad a > 0$	$\sqrt{\frac{\pi}{2}} \exp(-ak), \quad k > 0$

13.4 Laplace Transforms

	$f(t)$	$\bar{f}(s) = \int_0^\infty \exp(-st) f(t) dt$
1	$f^{(n)}(t)$	$s^n \bar{f}(s) - \sum_{r=0}^{n-1} s^{n-r-1} f^{(r)}(0)$
2	$\int_0^t f(t-\tau) g(\tau) d\tau$	$\bar{f}(s) \bar{g}(s)$
3	$t^n f(t)$	$(-1)^n \frac{d^n}{ds^n} \bar{f}(s)$
4	$f(t-a) H(t-a)$	$\exp(-as) \bar{f}(s)$
5	$t^n \quad (n = 0, 1, 2, 3, \dots)$	$\frac{n!}{s^{n+1}}$
6	e^{at}	$\frac{1}{s-a}$
7	$t^n e^{-at}$	$\frac{\Gamma(n+1)}{(s+a)^{n+1}}$
8	$t^a \quad (a > -1)$	$\frac{\Gamma(a+1)}{s^{a+1}}$
9	$e^{at} \cos bt$	$\frac{s-a}{(s-a)^2 + b^2}$
10	$e^{at} \sin bt$	$\frac{b}{(s-a)^2 + b^2}$
11	$\frac{1}{\sqrt{t}}$	$\sqrt{\frac{\pi}{s}}$
12	$2\sqrt{t}$	$\frac{1}{s} \sqrt{\frac{\pi}{s}}$

	$f(t)$	$\bar{f}(s) = \int_0^\infty \exp(-st) f(t) dt$
13	$t^{-1/2} \exp\left(-\frac{a}{t}\right)$	$\sqrt{\frac{\pi}{s}} \exp(-2\sqrt{as})$
14	$t^{-3/2} \exp\left(-\frac{a}{t}\right)$	$\sqrt{\frac{\pi}{a}} \exp(-2\sqrt{as})$
15	$\frac{1}{\sqrt{\pi t}} (1 + 2at) e^{at}$	$\frac{s}{(s-a)\sqrt{s-a}}$
16	$\frac{1}{2\sqrt{\pi t^3}} (e^{bt} - e^{at})$	$\sqrt{s-a} - \sqrt{s-b}$
17	$\exp(a^2 t) \operatorname{erf}(a\sqrt{t})$	$\frac{a}{\sqrt{s(s-a^2)}}$
18	$\exp(a^2 t) \operatorname{erfc}(a\sqrt{t})$	$\frac{1}{\sqrt{s}(\sqrt{s+a})}$
19	$\frac{1}{\sqrt{\pi t}} + a \exp(a^2 t) \operatorname{erf}(a\sqrt{t})$	$\frac{\sqrt{s}}{(s-a^2)}$
20	$\frac{1}{\sqrt{\pi t}} - a \exp(a^2 t) \operatorname{erfc}(a\sqrt{t})$	$\frac{1}{\sqrt{s+a}}$
21	$\frac{\exp(-at)}{\sqrt{b-a}} \operatorname{erf}\left(\sqrt{(b-a)t}\right)$	$\frac{1}{(s+a)\sqrt{s+b}}$
22	$\frac{1}{2} e^{i\omega t} \left[\exp(-\lambda z) \operatorname{erfc}\left(\zeta - \sqrt{i\omega t}\right) + \exp(\lambda z) \operatorname{erfc}\left(\zeta + \sqrt{i\omega t}\right) \right],$ where $\zeta = z/2\sqrt{\nu t}$, $\lambda = \sqrt{\frac{i\omega}{\nu}}$	$(s - i\omega)^{-1} \exp\left(-z\sqrt{\frac{s}{\nu}}\right)$
23	$\frac{1}{2} \left[\exp(-ab) \operatorname{erfc}\left(\frac{b-2at}{2\sqrt{t}}\right) + \exp(ab) \operatorname{erfc}\left(\frac{b+2at}{2\sqrt{t}}\right) \right]$	$\exp\left[-b(s+a^2)^{\frac{1}{2}}\right]$

	$f(t)$	$\bar{f}(s) = \int_0^\infty \exp(-st) f(t) dt$
24	$J_0(at)$	$(s^2 + a^2)^{-\frac{1}{2}}$
25	$I_0(at)$	$(s^2 - a^2)^{-\frac{1}{2}}$
26	$t^{\alpha-1} \exp(-at), \quad \alpha > 0$	$\Gamma(\alpha)(s+a)^{-\alpha}$
27	$t^{-1} J_\nu(at)$	$\nu^{-1} a^\nu (\sqrt{s^2 + a^2} + s)^{-\nu},$ $\operatorname{Re} \nu > -\frac{1}{2}$
28	$J_0(a\sqrt{t})$	$\frac{1}{s} \exp\left(-\frac{a^2}{4s}\right)$
29	$\left(\frac{2}{a}\right)^\nu t^{\nu/2} J_\nu(a\sqrt{t})$	$s^{-(\nu+1)} \exp\left(-\frac{a^2}{4s}\right),$ $\operatorname{Re} \nu > -\frac{1}{2}$
30	$\frac{a}{2t\sqrt{\pi t}} \exp\left(-\frac{a^2}{4t}\right)$	$\exp(-a\sqrt{s}), \quad a > 0$
31	$\frac{1}{\sqrt{\pi t}} \exp\left(-\frac{a^2}{4t}\right)$	$\frac{1}{\sqrt{s}} \exp(-a\sqrt{s}), \quad a \geq 0$
32	$\exp\left(-\frac{a^2 t^2}{4}\right)$	$\frac{\sqrt{\pi}}{a} \exp\left(\frac{s^2}{a^2}\right) \operatorname{erfc}\left(\frac{s}{a}\right),$ $a \geq 0$
33	$(t^2 - a^2)^{-\frac{1}{2}} H(t - a)$	$K_0(as), \quad a > 0$
34	$\delta^{(n)}(t - a), \quad n = 0, 1, \dots$	$s^n \exp(-as)$

	$f(t)$	$\bar{f}(s) = \int_0^\infty \exp(-st) f(t) dt$
35	$t^{m\alpha+\beta-1} E_{\alpha,\beta}^{(m)}(\pm at),$ $m = 0, 1, 2, \dots$	$\frac{m! s^{\alpha-\beta}}{(s^\alpha \mp a)^{m+1}}$
36	$\frac{\sqrt{\pi}}{\Gamma(\nu+\frac{1}{2})} \left(\frac{t}{2a}\right)^\nu J_\nu(at)$	$(s^2 + a^2)^{-(\nu+\frac{1}{2})},$ $\operatorname{Re} \nu > -\frac{1}{2}$
37	$\frac{1}{2} e^{-ct} \left[\exp(-a\sqrt{b-c}) \right.$ $\quad \times \operatorname{erfc} \left\{ \frac{a}{\sqrt{4t}} - \sqrt{(b-c)t} \right\}$ $\quad - \exp(a\sqrt{b-c})$ $\quad \times \operatorname{erfc} \left\{ \frac{a}{\sqrt{4t}} + \sqrt{(b-c)t} \right\} \Big]$	$\frac{\exp(-a\sqrt{s+b})}{(s+c)\sqrt{(s+b)}}$
38	$\frac{1}{2} e^{-ct} \left[\exp(-a\sqrt{b-c}) \right.$ $\quad \times \operatorname{erfc} \left\{ \frac{a}{\sqrt{4t}} - t\sqrt{b-c} \right\}$ $\quad - \exp(a\sqrt{b-c})$ $\quad \times \operatorname{erfc} \left\{ \frac{a}{\sqrt{4t}} + t\sqrt{b-c} \right\} \Big]$	$\frac{\exp(-a\sqrt{s+b})}{(s+c)}$
39	$e^{-bt} \left[\sqrt{\frac{4t}{\pi}} \exp\left(-\frac{a^2}{4t}\right) \right.$ $\quad \left. - a \operatorname{erfc}\left(\frac{a}{\sqrt{4t}}\right) \right]$	$\frac{\exp(-a\sqrt{s+b})}{(s+b)^{3/2}}$
40	$e^{-bt} \left[\left(t + \frac{1}{2}a^2\right) \operatorname{erfc}\left(\frac{a}{\sqrt{4t}}\right) \right.$ $\quad \left. - \sqrt{\frac{ta^2}{\pi}} \exp\left(-\frac{a^2}{4t}\right) \right]$	$\frac{\exp(-a\sqrt{s+b})}{(s+b)^2}$

13.5 Hankel Transforms

	$f(r)$	order n	$\tilde{f}_n(k) = \int_0^\infty r J_n(kr) f(r) dr$
1	$H(a-r)$	0	$\frac{a}{k} J_1(ak)$
2	$\exp(-ar)$	0	$a(a^2 + k^2)^{-\frac{3}{2}}$
3	$\frac{1}{r} \exp(-ar)$	0	$(a^2 + k^2)^{-\frac{1}{2}}$
4	$(a^2 - r^2) H(a-r)$	0	$\frac{4a}{k^3} J_1(ak) - \frac{2a^2}{k^2} J_0(ak)$
5	$a(a^2 + r^2)^{-\frac{3}{2}}$	0	$\exp(-ak)$
6	$\frac{1}{r} \cos(ar)$	0	$(k^2 - a^2)^{-\frac{1}{2}} H(k-a)$
7	$\frac{1}{r} \sin(ar)$	0	$(a^2 - k^2)^{-\frac{1}{2}} H(a-k)$
8	$\frac{1}{r^2} (1 - \cos ar)$	0	$\cosh^{-1}\left(\frac{a}{k}\right) H(a-k)$
9	$\frac{1}{r} J_1(ar)$	0	$\frac{1}{a} H(a-k), \quad a > 0$
10	$Y_0(ar)$	0	$\left(\frac{2}{\pi}\right) (a^2 - k^2)^{-1}$
11	$K_0(ar)$	0	$(a^2 + k^2)^{-1}$
12	$\frac{\delta(r)}{r}$	0	1

	$f(r)$	order n	$\tilde{f}_n(k) = \int_0^\infty r J_n(kr) f(r) dr$
13	$(r^2 + b^2)^{-\frac{1}{2}} \times \exp \left\{ -a (r^2 + b^2)^{\frac{1}{2}} \right\}$	0	$(k^2 + a^2)^{-\frac{1}{2}} \exp \left\{ -b (k^2 + a^2)^{\frac{1}{2}} \right\}$
14	$(r^2 + a^2)^{-\frac{1}{2}}$	0	$\frac{1}{k} \exp(-ak)$
15	$\exp(-ar)$	1	$k (a^2 + k^2)^{-3/2}$
16	$\frac{\sin ar}{r}$	1	$\frac{a H(k-a)}{k(k^2 - a^2)^{\frac{1}{2}}}$
17	$\frac{1}{r} \exp(-ar)$	1	$\frac{1}{k} \left[1 - \frac{a}{(k^2 + a^2)^{\frac{1}{2}}} \right]$
18	$\frac{1}{r^2} \exp(-ar)$	1	$\frac{1}{k} \left[(k^2 + a^2)^{\frac{1}{2}} - a \right]$
19	$r^n H(a - r)$	> -1	$\frac{1}{k} a^{n+1} J_{n+1}(ak)$
20	$r^n \exp(-ar), (\operatorname{Re} a > 0)$	> -1	$\frac{1}{\sqrt{\pi}} \frac{2^{n+1} \Gamma(n + \frac{3}{2}) a k^n}{(a^2 + k^2)^{n + \frac{3}{2}}}$
21	$r^n \exp(-ar^2)$	> -1	$\frac{k^n}{(2a)^{n+1}} \exp\left(-\frac{k^2}{4a}\right)$
22	r^{a-1}	> -1	$\frac{2a \Gamma[\frac{1}{2}(a+n+1)]}{k^{a+1} \Gamma[\frac{1}{2}(1-a+n)]}$

	$f(r)$	order n	$\tilde{f}_n(k) = \int_0^\infty r J_n(kr) f(r) dr$
23	$r^n (a^2 - r^2)^{m-n-1} \times H(a-r)$	> -1	$2^{m-n-1} \Gamma(m-n) a^m k^{n-m} J_m(ak)$
24	$r^m \exp(-r^2/a^2)$	> -1	$\frac{k^n a^{m+n+2}}{2^{n+1} \Gamma(n+1)} \Gamma\left(1 + \frac{m}{2} + \frac{n}{2}\right) \times {}_1F_1\left(1 + \frac{m}{2} + \frac{n}{2}; n+1; -\frac{1}{4}a^2k^2\right)$
25	$\frac{1}{r} J_{n+1}(ar)$	> -1	$k^n a^{-(n+1)} H(a-k), a > 0$
26	$r^n (a^2 - r^2)^m H(a-r),$ $m > -1$	> -1	$2^m a^n \Gamma(m+1) \left(\frac{a}{k}\right)^{m+1} \times J_{n+m+1}(ak)$
27	$\frac{1}{r^2} J_n(ar)$	$> \frac{1}{2}$	$\left\{ \begin{array}{l} \frac{1}{2n} \left(\frac{k}{a}\right)^n, \quad 0 < k \leq a \\ \frac{1}{2n} \left(\frac{a}{k}\right)^n, \quad a < k < \infty \end{array} \right\}$
28	$\frac{r^n}{(a^2+r^2)^{m+1}}, a > 0$	> -1	$\left(\frac{k}{2}\right)^m \frac{a^{n-m}}{\Gamma(m+1)} K_{n-m}(ak)$
29	$\exp(-p^2r^2) J_n(ar)$	> -1	$(2p^2)^{-1} \exp\left(-\frac{a^2+k^2}{4p^2}\right) I_n\left(\frac{ak}{2p^2}\right)$
30	$\frac{1}{r} \exp(-ar)$	> -1	$\frac{(k^2+a^2)^{\frac{1}{2}-a} - a^n}{k^n (k^2+a^2)^{\frac{1}{2}}}$
31	$\frac{r^n}{(r^2+a^2)^{n+1}}$	> -1	$\left(\frac{k}{2}\right)^n \frac{K_0(ak)}{\Gamma(n+1)}$
32	$\frac{r^n}{(a^2-r^2)^{n+\frac{1}{2}}} H(a-r)$	< 1	$\frac{1}{\sqrt{\pi}} \left(\frac{k}{2}\right)^n \Gamma\left(\frac{1}{2} - n\right) \left(\frac{\sin ak}{k}\right)$

	$f(r)$	order n	$\tilde{f}_n(k) = \int_0^\infty r J_n(kr) f(r) dr$
33	$f(ar)$	n	$\frac{1}{a^2} \tilde{f}_n\left(\frac{k}{a}\right)$
34	$r^{-1} \exp(-ar^2)$	1	$\frac{1}{k} \left[1 - \exp\left(-\frac{k^2}{4a}\right)\right]$
35	$r^{-1} \sin(ar^2), \quad a > 0$	1	$\frac{1}{k} \sin\left(\frac{k^2}{4a}\right)$
36	$r^{-1} \cos(ar^2), \quad a > 0$	1	$1 - \cos\left(\frac{k^2}{4a}\right)$
37	$\exp(-ar), \quad a > 0$	> -1	$\frac{(a+n\sqrt{k^2+a^2})}{(k^2+a^2)^{3/2}} \left(\frac{k}{a+\sqrt{a^2+k^2}}\right)^n$
38	$\exp(-ar^2) J_0(br)$	0	$\frac{a}{2} \exp\left(-\frac{k^2-b^2}{4a}\right) I_0\left(\frac{bk}{2a}\right)$
39	$\frac{H(a-r)}{\sqrt{a^2-r^2}}$	0	$\sqrt{\frac{a\pi}{2k}} J_{\frac{1}{2}}(ak), \quad a > 0$
40	$\frac{r^n H(a-r)}{\sqrt{a^2-r^2}}$	> -1	$\sqrt{\frac{\pi}{2k}} a^{n+\frac{1}{2}} J_{n+1}(ak), \quad a > 0$
41	$r^{-2} \sin r$	0	$\sin^{-1}\left(\frac{1}{k}\right), \quad (k > 1)$

13.6 Finite Hankel Transforms

	$f(r)$	order n	$\tilde{f}_n(k_i) = \int_0^a r J_n(r k_i) f(r) dr$
1	c , where c is a constant	0	$\left(\frac{ac}{k_i}\right) J_1(ak_i)$
2	$(a^2 - r^2)$	0	$\frac{4a}{k_i^3} J_1(ak_i)$
3	$(a^2 - r^2)^{-\frac{1}{2}}$	0	$k_i^{-1} \sin(ak_i)$
4	$\frac{J_0(\alpha r)}{J_0(\alpha a)}$	0	$-\frac{ak_i}{(\alpha^2 - k_i^2)} J_1(ak_i)$
5	$\frac{1}{r}$	1	$k_i^{-1} \{1 - J_0(ak_i)\}$
6	$r^{-1} (a^2 - r^2)^{-\frac{1}{2}}$	1	$\frac{(1 - \cos ak_i)}{(ak_i)}$
7	r^n	> -1	$\frac{a^{n+1}}{k_i} J_{n+1}(ak_i)$
8	$\frac{J_\nu(\alpha r)}{J_\nu(\alpha a)}$	> -1	$\frac{ak_i}{(\alpha^2 - k_i^2)} J'_\nu(ak_i)$
9	$r^{-n} (a^2 - r^2)^{-\frac{1}{2}}$	> -1	$\frac{\pi}{2} \left\{ J_{\frac{n}{2}}\left(\frac{ak_i}{2}\right) \right\}^2$
10	$r^n (a^2 - r^2)^{-(n+\frac{1}{2})}$	$< \frac{1}{2}$	$\frac{\Gamma(\frac{1}{2}-n)}{\sqrt{\pi} 2^n} k_i^{n-1} \sin(ak_i)$
11	$r^{n-1} (a^2 - r^2)^{n-\frac{1}{2}}$	$> -\frac{1}{2}$	$\frac{\sqrt{\pi}}{2} \Gamma\left(n + \frac{1}{2}\right) \left(\frac{2}{k_i}\right)^n a^{2n} J_n^2\left(\frac{ak_i}{2}\right)$